IoT & AI-BASED SMART AGRICULTURE PLATFORM FOR YIELD, DISEASE, IRRIGATION AND PEST MANAGEMENT

INDIVIDUAL THESIS

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ABSTRACT

Paddy cultivation in Sri Lanka depends heavily on irrigation, yet scheduling in many smallholder fields still follows fixed rotations, canal turns and farmer intuition rather than field evidence. This wastes water and exposes the crop to stress at critical growth stages. Existing smart-irrigation solutions are mostly threshold-based and reactive, or are designed for other crops and environments, and few combine soil sensing, weather data and growth-stage awareness for Sri Lankan paddy. This thesis presents the Paddy Irrigation Optimization component of an IoT and AI-based smart agriculture platform for yield, disease, irrigation and pest management. Its aim is a low-cost, stage-aware machine learning scheduler that recommends when to irrigate and how urgently, targeting a 30-50% reduction in water wastage. Following a design science and experimental evaluation approach, low-cost sensor nodes capture soil moisture, electrical conductivity, pH and temperature in pilot paddy plots in Negombo, while weather attributes and farmer-entered events complete the dataset. After validation, time alignment and growth-stage labelling, scikit-learn models (a threshold baseline, decision tree, random forest and gradient boosting) are compared on accuracy, interpretability and suitability for low-cost deployment. Recommendations are delivered through cloud APIs to a bilingual Sinhala-English mobile application with manual override and are shared with the yield, disease and pest components. The evaluation targets above 90% validation accuracy for irrigation-status prediction, measurable water savings against current farmer practice and at least 80% farmer satisfaction. [Insert final measured accuracy, water saving and satisfaction results after Yala 2026 validation.] The work contributes a stage-aware, locally validated scheduling approach, a reusable sensing-to-recommendation pipeline and an integration design linking irrigation to the wider platform.

Keywords: paddy irrigation, machine learning, Internet of Things, precision agriculture, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DOA	Department of Agriculture
EC	Electrical Conductivity
IoT	Internet of Things
LoRa	Long Range (low-power wide-area radio)
LSTM	Long Short-Term Memory
ML	Machine Learning
RRDI	Rice Research and Development Institute
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
WBS	Work Breakdown Structure
YOLO	You Only Look Once (object detector)

1 INTRODUCTION

1.1 Overall project context

Paddy cultivation is one of the most water-intensive agricultural activities in Sri Lanka, yet irrigation decisions in many fields still depend on fixed schedules, canal rotations and farmer intuition rather than real-time field evidence. At the same time, four recurring decision problems converge on the same paddy field: uncertainty about expected yield, disease outbreaks that are often identified only after symptoms appear, irrigation that follows routine rather than plot-level soil moisture and crop growth signals, and pest control that is reactive and costly. Standalone applications cannot solve such interconnected biological problems; a holistic platform in which data from the soil informs the response to the sky is required.

The IoT & AI-Based Smart Agriculture Platform for Yield, Disease, Irrigation and Pest Management (project code R26-IT-157) [1] addresses this need as one interoperable platform. It fuses sensor data, field images and expert knowledge to enable earlier and more practical decisions for smallholder paddy farmers, and delivers them as Sinhala and English alerts through shared APIs. The platform is developed as four linked components that share a common data backbone, so that data capture is not duplicated and each component can use the outputs of the others. Table 1.1 summarises the components and their owners; this thesis reports the irrigation component.

Table 1.1. Platform components and their owners

Component	Student	Core technique	Principal output
Yield forecasting	V.G.S.M. Siriwardhana (IT22189158)	XGBoost regressor on soil, weather and RRDI records	4-6 week yield outlook with confidence
Irrigation optimization (this thesis)	**J.M.S.U.I. Bandara (IT22269348)**	**Stage-aware ML scheduling (scikit-learn) with threshold baseline**	**Irrigation timing, urgency and expected water use**
Disease detection and risk forecasting	A.P.N. Satyana (IT22205506)	ResNet-50 CNN with LSTM on leaf images and weather sequences	7- and 14-day outbreak risk scores

Component	Student	Core technique	Principal output
Pest management	P.D.M. Kumarasinhe (IT22254566)	YOLOv8 trap-image counting with meteorological data	7-10 day larval hatching risk and treatment timing

1.2 Individual component context

The Paddy Irrigation Optimization component converts field sensing, weather information and growth-stage context into practical water-management advice for the farmer. Low-cost sensor nodes record soil moisture, electrical conductivity (EC), pH and temperature; weather attributes such as rainfall, humidity and short-range forecast are retrieved from connected data services; and farmer-entered events such as irrigation dates and observed crop stage add context. A machine learning (ML) scheduler then predicts whether irrigation is required now, should be delayed or should be monitored, together with its urgency and expected water use.

The component is not isolated. Its outputs support yield estimation by supplying irrigation history, stress trends and stage data, and they affect disease and pest risk indirectly by changing soil moisture and field humidity. In the other direction, soil moisture provides wetness context for the disease component's LSTM predictions, outbreak alerts can trigger spray-window-aware watering adjustments, and trap microclimate and risk scores feed the yield forecast. The pilot is carried out in paddy plots in the Negombo area across the Maha and Yala seasons, with validation planned during Yala 2026.

1.3 Problem statement

Irrigation in many smallholder paddy fields is scheduled by routine rather than by evidence. Flood irrigation and canal rotations deliver water on a calendar, so the amount delivered often differs from the amount the crop actually needs, particularly when rainfall shifts unexpectedly during the growing season. The consequences are twofold. Delayed or insufficient irrigation causes water stress at sensitive stages such as tillering and flowering, which can reduce yield. Excess water wastes a scarce resource, reduces nutrient efficiency and can create field conditions that favour disease.

Smallholder farmers are particularly affected because they manage limited land and have little margin for water waste, late irrigation or yield loss. Sensor-based threshold systems improve on fixed schedules but are reactive: a valve opens when the soil is dry, yet the system does not learn how the crop's water demand changes with growth stage and season. The problem addressed in this thesis is therefore the absence of an affordable, field-validated irrigation scheduler that anticipates paddy water demand using soil, weather and growth-stage information.

1.4 Research gap

Prior work in smart irrigation shows that sensor-driven control can reduce waste, but much of the literature is either threshold-based or designed for crops and production environments other than Sri Lankan paddy [2], [3], [4]. Mobile and IoT tools for rice fields exist but offer limited growth-stage reasoning and shallow ML [5], while ML-based and model-predictive scheduling is demonstrated in settings that are not tailored to paddy conditions in Sri Lanka [6].

The research gap is therefore the absence of a low-cost, Sri Lankan paddy-specific irrigation optimizer that combines soil sensing, weather data, growth-stage awareness and machine learning in a single operational pipeline, and that is validated against current farmer practice in the Maha and Yala seasons. Chapter 2 develops this gap in detail.

1.5 Aim and objectives

Aim. To optimize paddy irrigation through a stage-aware machine learning scheduler that reduces water wastage by 30-50% while supporting healthier crop growth and integrating with the wider smart agriculture platform.

The aim is broken down into four specific objectives, each with a measurable indicator (Table 1.2).

Table 1.2. Specific objectives and indicators

ID	Objective	Indicator
SO1	Deploy and calibrate field sensors for soil and environmental monitoring	Stable pilot data stream available for model input

ID	Objective	Indicator
SO2	Develop a stage-aware ML model for irrigation scheduling	Validation accuracy above 90% for irrigation-status prediction
SO3	Integrate the scheduler with cloud APIs and the mobile app	Alerts and history logs are operational
SO4	Validate the component in field conditions	Water savings and user feedback measured against current practice

1.6 Research questions or hypotheses

The study is guided by the following research questions and the hypotheses that answer them.

1. **RQ1.** How accurately can a stage-aware ML model predict the irrigation status (irrigate now, delay or monitor) from soil, weather and growth-stage features compared with a single-threshold baseline?
2. **RQ2.** How much water can be saved relative to current farmer practice while keeping crop growth healthy?
3. **RQ3.** Can irrigation recommendations be delivered reliably and usably to smallholder farmers through a bilingual mobile application with a safe fallback when sensors fail?

- **H1.** The stage-aware ML scheduler achieves higher validation accuracy than the threshold baseline and reaches at least 90% accuracy for irrigation-status prediction.
- **H2.** Following the scheduler's recommendations reduces water use by 30-50% relative to current farmer practice without visible crop stress.
- **H3.** At least 80% of participating farmers report satisfaction with the recommendations and the application.

1.7 Scope limitations and assumptions

Scope. The component covers sensing, data processing, ML scheduling, cloud integration and bilingual mobile delivery for paddy in pilot plots in Negombo. Validation takes place during the Yala 2026 season using a compact pilot kit of three

sensor nodes. During the pilot, recommendations are presented as guidance; any direct actuation of valves must be supervised and include a manual override.

Limitations. The study does not address crops other than paddy, large-scale commercial networks, fully autonomous valve control without supervision, or the yield, disease and pest models, which belong to the other components. The small number of nodes and a single pilot area limit how widely the results can be generalised.

Assumptions. Sensors can be calibrated and maintained during the season; farmers enter irrigation events and observed crop stage through the application; connectivity may be intermittent, so readings are buffered locally and synchronised when the network returns; and weather data services are available at the pilot interval.

1.8 Individual contribution

The author is responsible for the irrigation component of the platform. The individual contribution comprises:

- design and calibration of the soil and environmental sensing stack (Arduino/ESP32 controller with DFRobot-compatible probes) and its data-validation rules;
- the data pipeline, including time alignment, missing-value handling, outlier detection, smoothing and growth-stage labelling;
- engineering of stage-aware features and the training, comparison and selection of ML scheduling models in scikit-learn;
- the recommendation service, cloud APIs and alert logic, including fallback alerts for sensor faults and a farmer manual override;
- integration with the bilingual Sinhala-English mobile application and with the yield, disease and pest components; and
- the design and execution of the field validation against current farmer practice.

The yield forecasting, disease detection and pest management components are the work of other team members and are described here only where they interact with irrigation.

1.9 Thesis structure

Chapter 2 reviews the literature, existing systems and the research gap. Chapter 3 describes the research methodology, including data, procedure, evaluation measures and ethics. Chapter 4 presents the requirements and the design of the component within the complete solution. Chapter 5 covers implementation, integration and testing. Chapter 6 reports the results and validation, and Chapter 7 discusses them against the literature, together with contributions, limitations and implications. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the thesis and outlines future directions. References, a glossary and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter is a focused narrative review rather than a systematic review. It draws on the sources identified at the proposal stage, which cover low-cost IoT irrigation, mobile delivery of ricefield data, ML-based and model-predictive irrigation scheduling, threshold-based decision support and AI-based water optimization, and extends them with foundational references for the methods used in this thesis. Sources were selected for their relevance to sensor-driven or ML-based irrigation scheduling, with preference for work on rice or comparable field crops published between 2018 and 2025.

The review has three purposes: to establish the domain and theoretical foundations of paddy water management and supervised learning; to summarise what related research and existing systems offer; and to identify the gap that this component addresses.

2.2 Domain and theoretical foundations

Paddy water demand. Rice water demand is not constant over the season. Crop water requirement guidelines describe how evapotranspiration and crop coefficients change with development stage [7]. For the purposes of this thesis, four growth stages are used: tillering, panicle initiation, flowering and grain filling. Water stress is most damaging at tillering and flowering, whereas excess water at other stages reduces nutrient efficiency and promotes disease-conducive field conditions (Figure 2.1). A generic moisture threshold treats all stages alike and is therefore fundamentally limited.

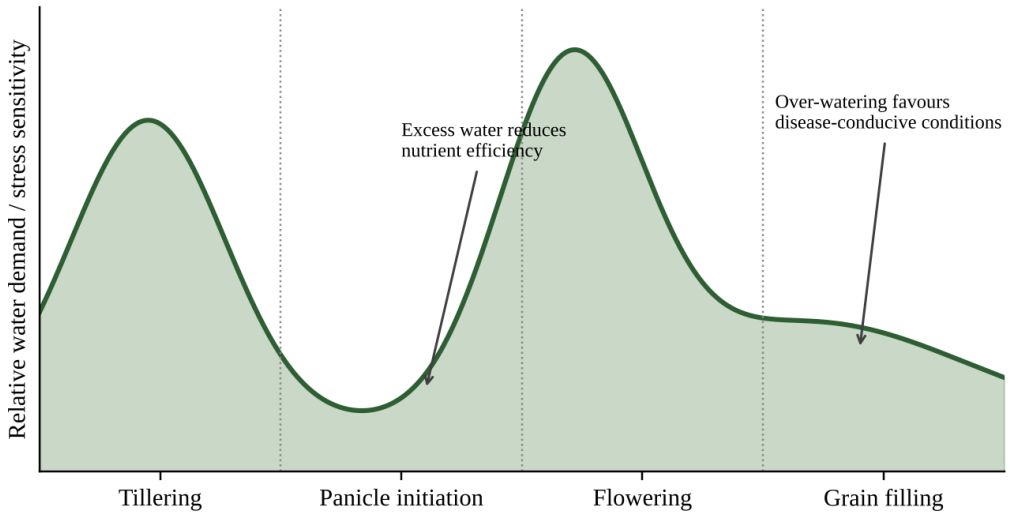


Figure 2.1. Conceptual stage-aware water demand across paddy growth stages

Sri Lankan seasons. Paddy is cultivated mainly in two seasons: Maha, the main season associated with the north-east monsoon, and Yala, the minor season. Rainfall patterns and irrigation availability differ between them, so a scheduler that is trained on one season may drift in the other. Seasonal context is therefore treated as an input feature and as a retraining trigger.

Sensing. Soil moisture indicates plant-available water, EC indicates dissolved salts and nutrient status, pH affects nutrient availability, and temperature influences evapotranspiration and sensor behaviour. Together with rainfall, humidity and short-range forecasts they describe the water balance of a plot at a given moment.

Supervised learning for scheduling. Irrigation status can be framed as a classification problem (irrigate now, delay or monitor) and, where feasible, as a regression problem for duration or priority. Decision trees give transparent rules; random forests average many trees to reduce variance [8]; and gradient boosting builds trees sequentially to correct earlier errors [9]. These algorithms perform well on small tabular datasets, are inexpensive to run and are available in scikit-learn [10], which makes them suitable for low-cost deployment.

Research approach. The work follows design science, in which a useful artefact is built and evaluated iteratively against a real problem [11].

2.3 Related research

Table 2.1 summarises the studies that are most relevant to this component. Patil et al. [2] demonstrate practical, affordable irrigation scheduling that combines low-cost ML and IoT hardware, but for a generic crop focus rather than stage-specific paddy. Rahman et al. [5] show how soil moisture in a ricefield can be monitored with IoT hardware and delivered to an Android application, but with limited growth-stage reasoning and weak ML depth. Li et al. [6] combine machine learning with model-predictive control for irrigation scheduling and report 6-23% savings in comparative settings, though not for Sri Lankan paddy conditions.

Attanasio et al. [3] describe a decision support system with threshold-based valve control, which is useful for reactive irrigation but does not learn stage-wise water demand. Mishra et al. [4] confirm the value of low-cost sensing and automation in an IoT smart irrigation architecture, while noting the need for crop-specific adaptation and local validation. Practitioner and regional sources on AI in rice production [12] and AI-based water optimization [13] reinforce the interest in this direction but do not provide a paddy-specific, stage-aware scheduler for Sri Lanka.

Table 2.1. Literature summary

Source	Approach	Finding	Gap
Patil et al. [2]	Low-cost ML and IoT irrigation scheduling	Demonstrates practical scheduling with affordable hardware	Generic crop focus; not stage-specific for paddy
Rahman et al. [5]	IoT soil monitoring with Android support for rice	Shows mobile delivery of ricefield data	Limited growth-stage reasoning; weak ML depth
Li et al. [6]	ML and model-predictive irrigation scheduling	Reported 6-23% savings in comparative settings	Not designed for Sri Lankan paddy conditions
Attanasio et al. [3]	Decision support with threshold-based valve control	Useful for reactive irrigation control	Does not learn stage-wise water demand
Mishra et al. [4]	IoT-based smart irrigation architecture	Confirms value of low-cost sensing and automation	Needs crop-specific adaptation and local validation

2.4 Existing systems products and techniques

In practice, three families of approach are available to a Sri Lankan smallholder, plus the approach proposed here (Table 2.2). Flood irrigation and canal rotations are simple and familiar but waste water and offer no field-level precision. Threshold sensor systems open a valve when soil moisture crosses a limit; the project's baseline review suggests that they can save roughly 20-30% of water, but their logic is reactive and handles crop stage and season poorly. Generic ML scheduling uses weather and soil variables for predictive decision support, but is often not tailored to Sri Lankan paddy cultivation.

Table 2.2. Comparison of existing irrigation approaches and the proposed component

Approach	Data	Strength	Limitation
Flood irrigation and canal rotation	Manual observation, calendar	Simple and familiar for farmers	High water wastage; no field-level precision
Threshold sensor systems	Soil moisture thresholds	Can save roughly 20-30% of water	Reactive logic; weak handling of crop stage and season
Generic ML scheduling	Weather and soil variables	Predictive decision support	Often not tailored to Sri Lankan paddy cultivation
Proposed stage-aware ML scheduler	**Soil, weather, growth stage, farmer events**	**Learns Maha/Yala and stage-specific demand; low-cost; bilingual delivery**	**Requires local data and seasonal retraining**

2.5 Gap synthesis and conceptual framework

Three observations follow from the review. First, sensor-driven irrigation reduces waste, but threshold logic is reactive and does not capture how paddy water demand changes with growth stage. Second, ML-based scheduling exists but is rarely designed for, or validated on, Sri Lankan paddy in the Maha and Yala seasons. Third, solutions that reach smallholders must be affordable, bilingual and safe when sensors fail, and should connect to other decisions on the same field.

These observations give the conceptual framework of the thesis. Field and weather data are captured and validated; growth-stage context is attached; stage-aware

features are engineered; an ML scheduler predicts irrigation status, urgency and expected water use; the recommendation is delivered to the farmer, who can act on it or override it; and the logged decision and outcome are stored to refine the model. The same data and decisions are shared with the yield, disease and pest components. The framework is operationalised in Chapters 3 and 4.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a design science and experimental evaluation approach [11]. The component is designed as a practical decision-support artefact that moves from requirement gathering through sensing, modelling and integration to field validation. The overall method is to capture field and weather data, preprocess them and engineer growth-stage features, train and compare ML models, deploy the best-performing model through cloud APIs, and validate its usefulness under pilot conditions. Figure 3.1 shows the irrigation decision workflow that the artefact implements in each reading cycle.

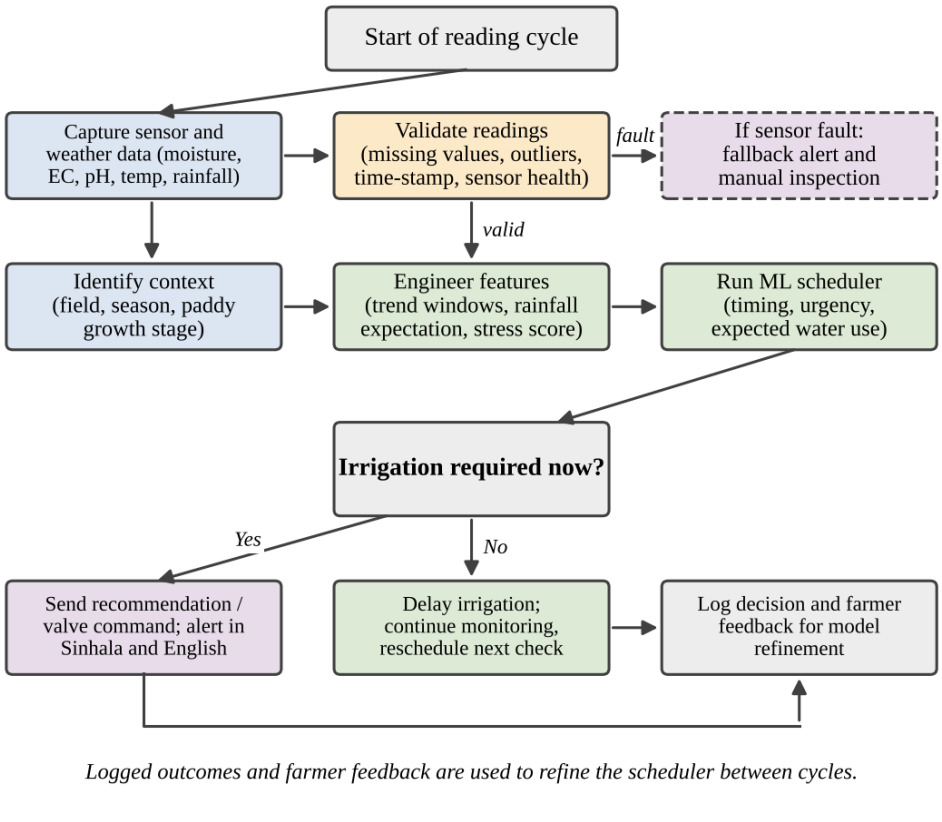


Figure 3.1. Irrigation decision workflow of the paddy optimization component

Evaluation is two-stage: offline validation with historical and pilot data to select and tune the model, followed by field evaluation against current farmer practice during Yala 2026.

3.2 Data participants or study materials

Data are collected from sensor nodes installed in pilot paddy plots in Negombo. The sensor package records soil moisture, EC, pH and temperature. Weather attributes such as rainfall, humidity and short-range forecast are retrieved from connected data services, and farmer-entered events such as irrigation date, field condition and observed crop stage are added through the mobile interface to strengthen contextual interpretation. Table 3.1 lists the variables.

Table 3.1. Variables used by the scheduler

Variable	Source	Role
Soil moisture	Field sensor node	Primary indicator of plant-available water; trend features
Electrical conductivity (EC)	Field sensor node	Salinity and nutrient context; data-quality check
Soil pH	Field sensor node	Nutrient availability context
Soil / ambient temperature	Field sensor node	Evapotranspiration context; sensor-quality check
Rainfall, humidity, short-range forecast	Weather data service	Rainfall expectation; delay or reduce irrigation
Growth stage (tillering, panicle initiation, flowering, grain filling)	Farmer entry and season calendar	Stage-specific water demand
Irrigation date and field condition	Farmer entry via mobile app	Interval since last irrigation; labels and feedback
Season (Maha / Yala)	Calendar	Seasonal context; drift monitoring

Participants are smallholder paddy farmers in the Negombo pilot area, with agricultural officers and RRDI stakeholders as secondary users of the history and data. The pilot kit consists of three field-ready node bundles, which allows comparison across field points. Participation is voluntary and based on informed consent.

3.3 Experimental or study procedure

The procedure has seven steps.

1. **Requirements and system design.** Stakeholder needs are elicited and translated into requirements and an architecture (Chapter 4).
2. **Sensor setup and calibration.** Nodes are built, calibrated and deployed in pilot plots, and a stable data stream is established.
3. **Data collection.** Sensor, weather and farmer-event data are collected through the Yala 2026 season, supplemented by available historical data.
4. **Preprocessing.** Time-stamps are aligned, missing values handled, outliers detected and noisy readings smoothed; each record is labelled with the growth stage (tillering, panicle initiation, flowering or grain filling).
5. **Feature engineering.** Moisture trend, interval since last irrigation, rainfall expectation, a water-stress score, stage factors and seasonal context are derived.
6. **Model development.** Candidate models (Table 3.2) are trained and compared. The output is framed as an irrigation recommendation class (irrigate now, delay or monitor) and, where feasible, a duration or priority score.
7. **Field evaluation.** The selected model is deployed through cloud APIs and evaluated against current farmer practice (Section 3.4).

Table 3.2. Candidate scheduling models

Model	Role	Rationale
Threshold rule	Baseline	Mimics existing reactive sensor systems; reference for improvement
Decision tree	Interpretable ML	Transparent rules that agronomists can inspect
Random forest [8]	Ensemble ML	Robust to noise and small datasets
Gradient boosting [9]	Ensemble ML	High accuracy on tabular data

Model selection depends on validation accuracy, interpretability, robustness and suitability for low-cost deployment. To avoid information leakage from correlated readings, validation uses time-aware splits (earlier periods for training, later periods for validation) in addition to cross-validation.

3.4 Measures and evaluation metrics

The main evaluation indicators are scheduling accuracy, application response time, data availability and water savings, complemented by farmer satisfaction (Table 3.3).

Table 3.3. Evaluation measures and targets

Measure	Definition	Target
Scheduling accuracy	Share of cycles in which the predicted irrigation status matches the reference status; precision, recall and F1 for the 'irrigate now' class are also reported	Above 90% validation accuracy
Water savings	Reduction in water used per plot relative to current farmer practice over the evaluation period	30-50%
Application response time	Load time of common status screens	Under 2 seconds
Data availability	Share of expected readings that reach the cloud within the pilot interval	[define target with supervisor]
Farmer satisfaction	Share of participating farmers reporting satisfaction in a short survey	At least 80%

3.5 Data analysis

Candidate models are compared on the validation partitions using accuracy, precision, recall and F1, with confusion matrices used to inspect errors on the 'irrigate now' class, since missed irrigation is the costliest error. The selected model is compared with the threshold baseline to answer RQ1 and test H1. Water use in recommendation-following plots is compared with that in current-practice plots, using paired or non-parametric comparisons as the sample size allows, to answer RQ2 and test H2. Survey responses and system logs (response times, synchronisation success, alert delivery) are summarised descriptively to answer RQ3 and test H3. Seasonal drift is monitored by tracking validation metrics across Maha and Yala data.

3.6 Ethics privacy security and responsible AI

Farmer participation is voluntary and based on informed consent. Field identifiers in the dataset are coded to avoid exposing personally identifiable information in

exported datasets or demonstrations. Sensor data, weather feeds and mobile interactions are used only for research and pilot evaluation within the project scope, and administrative access to the cloud database is restricted to authorised team members, consistent with Sri Lanka's Personal Data Protection Act, No. 9 of 2022.

Responsible AI is addressed in three ways. Recommendations are presented as guided decisions with transparent alert messages rather than unexplained commands. Any direct actuation is supervised and includes a manual override, preventing crop stress or over-irrigation caused by sensor failure. Finally, a fallback alert requests manual inspection when readings are missing or abnormal, so the system never fails silently.

3.7 Reproducibility and research artefacts

To support reproducibility, the following artefacts are maintained: versioned source code for firmware, data pipeline, models and application [insert repository link]; a data dictionary describing each variable, unit and sampling interval; the training and validation configuration (feature list, split definition, random seeds and model hyperparameters); and versioned model files with their validation reports. Raw farmer-linked data are shared only in coded form.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

User requirements were derived from the needs of farmers, agricultural officers, RRDI stakeholders and the project team. Farmers need a simple recommendation rather than raw sensor values; officers need visibility of irrigation history and field condition; researchers need clean data for validation; and the project team needs a secure, maintainable module that can scale beyond a single pilot field (Table 4.1).

Table 4.1. Stakeholders and their needs

Stakeholder	Need
Farmers	Receive timely irrigation recommendations, view alerts, record irrigation actions
Agricultural officers / extension staff	Monitor field history, compare advisory patterns, support adoption
RRDI / research stakeholders	Review data quality, seasonal trends and pilot outcomes
Project team / administrator	Maintain sensors, APIs, models and application services

The resulting requirements and their acceptance criteria are listed in Table 4.2.

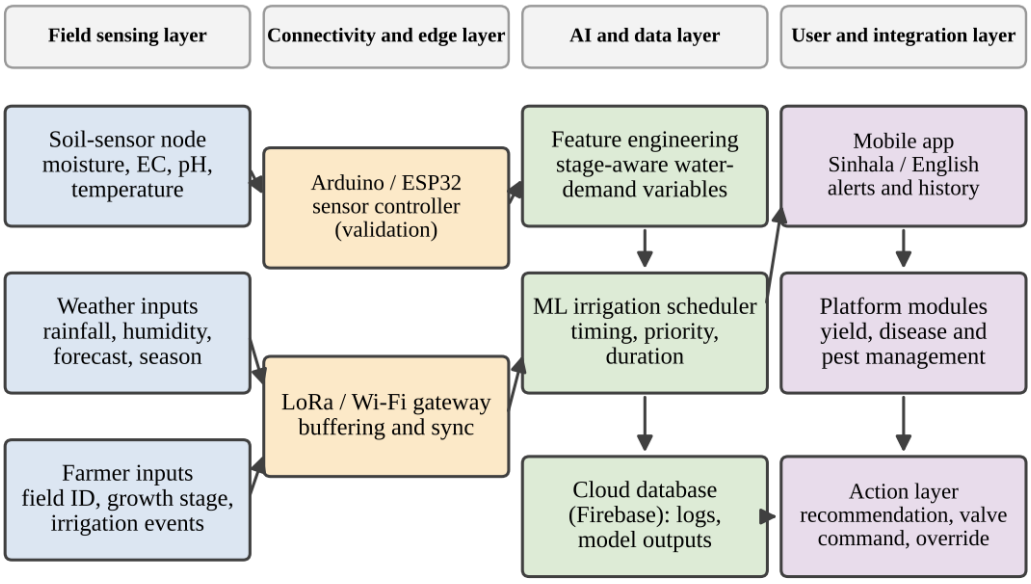
Table 4.2. Requirements and acceptance criteria

Category	Requirement	Acceptance criterion
Functional	Monitor soil and weather inputs from field nodes	Sensor readings synchronised to cloud within the pilot interval
Functional	Predict irrigation requirement and urgency	Recommendation generated after a validated input cycle
Functional	Send alerts and keep irrigation history	Farmer receives advisory status and can view past records
Performance	Return app responses quickly	Common status screens load in under 2 seconds
Usability	Provide a bilingual mobile interface	Key screens available in Sinhala and English
Reliability	Detect missing or abnormal sensor readings	System raises a fallback alert instead of failing silently

Category	Requirement	Acceptance criterion
Security	Protect cloud data and API access	Authenticated access and role-based permissions applied

4.2 Component architecture within the complete solution

The high-level architecture is layered (Figure 4.1). At the field level, sensor nodes capture soil and environmental observations. A controller and gateway layer performs basic validation and transmits readings to the cloud. The ML scheduler converts the data into irrigation recommendations, which are published to the mobile application and shared with the yield, disease and pest-management modules.



Stage-aware irrigation recommendations flow to the bilingual farmer app and the wider platform; logs and farmer feedback return to the cloud database for model refinement.

Figure 4.1. High-level architecture of the paddy irrigation optimization component

At platform level the design has three tiers: an edge tier of low-power, offline-tolerant nodes (Arduino and ESP32 for irrigation); a cloud tier with shared API services and a database for readings, logs and model outputs; and a delivery tier of authenticated HTTPS endpoints and role-based dashboards for farmers and agronomists.

This design supports modularity. The irrigation component remains responsible for water scheduling, while other modules consume irrigation history, stress indicators and microclimate context for their own predictions (Figure 4.2). The result is a connected agriculture platform rather than a standalone irrigation timer.

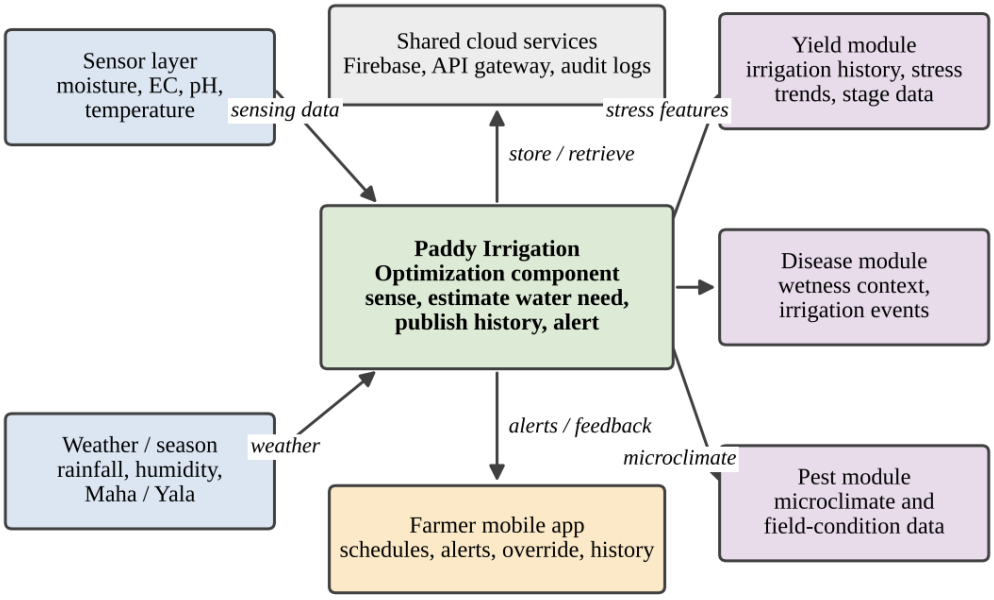


Figure 4.2. Interactions between the irrigation component and the rest of the platform

4.3 Design alternatives and decisions

Table 4.3 records the main design decisions, the alternatives considered and the reasons for the choices made.

Table 4.3. Design alternatives and decisions

Decision	Alternatives	Choice and rationale
Scheduling logic	Fixed schedule; threshold rules; ML scheduler	ML scheduler with a threshold baseline: learns stage- and season-specific demand and can be compared against reactive logic
Model family	Deep neural network; tree-based ensembles	Tree-based ensembles: suited to small tabular datasets, interpretable and cheap to run
Control mode	Autonomous valve control; advisory with manual override	Advisory with manual override during the pilot: avoids crop stress from sensor failure

Decision	Alternatives	Choice and rationale
Connectivity	Direct cloud link; LoRa or Wi-Fi gateway with local buffering	Gateway with buffering: prevents data loss during connectivity drops
Cloud service	Self-hosted database; managed real-time database (Firebase)	Firebase for the pilot: real-time synchronisation with the mobile app and low set-up cost
Hardware	Proprietary logger; Arduino/ESP32 with DFRobot-compatible probes	Off-the-shelf Arduino/ESP32 stack: low cost, widely available, staged funding possible

4.4 Detailed design

Sensing node. Each node combines DFRobot-compatible probes for soil moisture, EC and pH with a temperature sensor, connected to an Arduino or ESP32 controller. The controller time-stamps readings, checks them for missing or implausible values and sensor health, and buffers them locally when the network is unavailable.

Validation and fallback. Readings that fail validation are not passed to the scheduler. Instead, the system raises a fallback alert and requests manual inspection or sensor maintenance, so a faulty sensor cannot silently cause a wrong irrigation decision.

Feature engineering and scheduler. Validated readings are combined with weather data and growth-stage context to produce features: moisture trend windows, interval since last irrigation, rainfall expectation, a water-stress score and stage factors. The scheduler outputs an irrigation recommendation class (irrigate now, delay or monitor) and, where feasible, a duration or priority score and expected water use.

Delivery and feedback. Recommendations are published through cloud APIs to the mobile application, which shows status, recommendation, expected water use, history and a language switch between Sinhala and English. The farmer can accept or override a recommendation; each decision and outcome is logged and stored for model refinement, closing the feedback loop shown in Figure 3.1.

Data model. The cloud database holds four logical collections: sensor readings (per node and time-stamp), farmer events (irrigation, field condition, observed stage),

recommendations (class, urgency, expected water use, model version) and feedback (farmer action and outcome).

4.5 Standards and professional practice

The work follows professional practice in four respects. Sources are cited consistently in IEEE style. Software is developed under version control with separate configuration for model parameters. Personal and field data are handled in line with informed consent, coded identifiers and role-based access (Section 3.6). Finally, open-source libraries are used under their licences, and safety is treated as a design requirement: advisory recommendations, manual override and fallback alerts are built into the system rather than added afterwards.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1 lists the tools and platforms used, with the justification for each.

Table 5.1. Tools and platforms

Tool	Purpose	Justification
Arduino / DFRobot-compatible sensor stack	Field sensing and edge control	Low-cost, widely available, suitable for pilot sensor nodes
ESP32 controller	Edge processing and Wi-Fi connectivity	Low power with built-in Wi-Fi
LoRa / Wi-Fi gateway	Buffering and synchronisation	Tolerates intermittent connectivity
Python and scikit-learn [10]	Model development and evaluation	Supports interpretable ML baselines suitable for deployment
Firebase	Cloud synchronisation, API integration, alert logging	Provides real-time synchronisation for the mobile application
Android / Flutter mobile front end	Farmer-facing application layer	Supports Sinhala-English alerts and field history views

5.2 Component implementation

Edge firmware. The node firmware reads the sensors at the configured interval [insert sampling interval], applies range and sensor-health checks, time-stamps each record and transmits it through the gateway. If the link is unavailable, readings are stored locally and retried when the network returns.

Data pipeline. A Python pipeline aligns time-stamps across sensor, weather and farmer-event sources, handles missing values, flags outliers, smooths noisy readings and assigns the growth-stage label. It then computes the engineered features described in Section 4.4.

Model training. The threshold baseline, decision tree, random forest and gradient boosting models are trained with scikit-learn using time-aware splits and cross-

validation. The selected model [insert selected model and key hyperparameters] is serialised with a version identifier and its validation report.

Recommendation service and alerts. The recommendation service loads the current model, scores each validated cycle and writes the recommendation, urgency and expected water use to the cloud database. Alert logic raises a Sinhala-English notification when irrigation is required and a fallback alert when readings are missing or abnormal.

Mobile application. The application, built with Android/Flutter, synchronises with Firebase and presents the current recommendation, field history and manual-override controls in Sinhala and English.

5.3 Integration with the other components

The irrigation component exchanges data with the other platform modules through shared APIs, as summarised in Table 5.2. This avoids duplicated data capture and lets each module use context that it cannot sense itself.

Table 5.2. Interfaces with other platform components

Interface	Direction	Content	Purpose
Irrigation history and stress trends	Irrigation to yield	Irrigation events, water-stress indicators, stage data	Input features for the yield forecast
Wetness context	Irrigation to disease	Soil moisture and irrigation events	Wetness context for LSTM risk prediction
Spray-window flag	Disease to irrigation	Outbreak alert and treatment window	Adjust watering around spray windows
Microclimate history	Irrigation to pest	Field humidity and soil condition	Context for pest risk scoring
Schedule API	Irrigation to mobile app	Recommendation, urgency, expected water use	Farmer-facing alerts and history

5.4 Verification and software testing

Verification is organised in levels, from individual functions to the complete pilot (Table 5.3).

Table 5.3. Verification and testing plan

Level	Focus	Example checks	Pass criterion
Unit	Pipeline and feature functions	Time alignment, missing-value handling, stage labelling	Expected outputs on test fixtures
Sensor and edge	Validation and fallback	Missing, frozen or out-of-range readings; connectivity loss and recovery	Fallback alert raised; buffered data synchronised
Model	Offline validation	Time-aware validation against the threshold baseline	Meets accuracy target; improves on baseline
Integration	APIs and cloud sync	Reading to recommendation to app; exchange with other modules	Synchronised within the pilot interval
Usability and performance	Mobile application	Sinhala and English screens; load time of status screens	Key screens bilingual; under 2 seconds
Security	Access control	Authenticated access; role-based permissions	Unauthorised access refused

5.5 Deployment and operation

The pilot is deployed as three field-ready node bundles in Negombo paddy plots, each including probes, controller and enclosure, with a gateway providing buffering and synchronisation to the cloud. During operation, the project team monitors data availability and sensor health; faults raise manual-inspection requests rather than silent failures. Recommendations are advisory, direct actuation is supervised and always allows manual override, and the model is retrained with seasonal data when validation metrics drift between Maha and Yala.

6 RESULTS AND VALIDATION

6.1 Research results

[Note - delete before submission: Results must come from the author's own experiments. Replace every highlighted placeholder in this chapter with measured values from the Yala 2026 pilot and the offline model comparison, and delete this note.]

This section reports the offline comparison of the candidate models against the threshold baseline on the validation partitions (RQ1, H1). The target is a validation accuracy above 90% for irrigation-status prediction. Table 6.1 reports the results.

Table 6.1. Model comparison on the validation data

Model	Accuracy	Precision ('irrigate now')	Recall ('irrigate now')	F1
Threshold baseline	[insert]	[insert]	[insert]	[insert]
Decision tree	[insert]	[insert]	[insert]	[insert]
Random forest	[insert]	[insert]	[insert]	[insert]
Gradient boosting	[insert]	[insert]	[insert]	[insert]

[Describe which model was selected and why, including the feature importance or rule structure that supports its interpretability, and add the confusion matrix as a figure.]

Water savings measured against current farmer practice (RQ2, H2) were [insert measured water use per plot and percentage saving for each pilot plot and the average].

6.2 Product and technical results

The technical results cover the acceptance criteria set in Table 4.2. [Report measured status-screen load times, synchronisation success rate within the pilot interval, fallback-alert behaviour during sensor or connectivity faults, and results of the security checks.]

6.3 User or stakeholder validation

Farmers participating in the pilot completed a short satisfaction survey on the clarity of recommendations, usefulness of the alerts and ease of using the Sinhala-English interface (RQ3, H3). The target is at least 80% satisfaction. [Insert number of respondents, satisfaction percentage and representative comments from farmers and agricultural officers.]

6.4 Integration results

[Report whether irrigation history, wetness context and microclimate data were successfully exchanged with the yield, disease and pest components (Table 5.2), including any observed effect, such as spray-window-aware watering adjustments.]

6.5 Summary of findings

Table 6.2 compares the targets set in Chapter 1 with the measured outcomes.

Table 6.2. Targets and measured outcomes

Indicator	Target	Measured
Validation accuracy for irrigation-status prediction	Above 90%	[insert]
Water savings against current practice	30-50%	[insert]
Farmer satisfaction	At least 80%	[insert]
Common status screen load time	Under 2 seconds	[insert]

7 DISCUSSION

7.1 Interpretation of findings

[Note - delete before submission: Complete this section once Chapter 6 contains measured results. The paragraphs below give the structure to follow.]

The interpretation should answer the three research questions in turn. For RQ1, it should explain why the selected model did or did not outperform the threshold baseline, linking differences to growth-stage features, rainfall expectation and interval since last irrigation. For RQ2, it should relate the measured water saving to the 30-50% target and to factors such as rainfall during the pilot. For RQ3, it should interpret farmer satisfaction and response-time results in terms of bilingual delivery and the manual-override design. [Add interpretation of the measured results.]

7.2 Comparison with literature baselines and existing solutions

The literature offers several reference points. Li et al. [6] report savings of 6-23% with ML and model-predictive scheduling in comparative settings, and the project's baseline review suggests roughly 20-30% for threshold sensor systems. The 30-50% target of this component is therefore more ambitious than these baselines, which is why it is tested against current farmer practice in the same fields rather than only against other systems. Unlike the studies in Table 2.1, the component is designed specifically for Sri Lankan paddy, uses growth-stage and seasonal context, and is delivered bilingually with manual override. [Compare the measured savings and accuracy with these baselines.]

7.3 Research contribution

The research contribution is a stage-aware approach to irrigation scheduling for Sri Lankan paddy that combines low-cost soil sensing, weather data and growth-stage features in an ML scheduler, together with an evaluation design that compares it with a reactive threshold baseline and with current farmer practice across the Maha and Yala seasons.

7.4 Product contribution

The product contribution is a working sensing-to-recommendation pipeline: edge nodes with local buffering and fallback alerts, a scheduling service, cloud APIs and a bilingual mobile application with manual override. It is built from off-the-shelf components within a pilot budget of LKR 80,000 and integrates with the yield, disease and pest components of the platform.

7.5 Limitations and threats to validity

- **Internal validity.** Sensor drift or poor calibration can bias readings; rainfall and other weather differences between plots can confound water-saving comparisons; and the irrigation-status reference labels depend on how they are defined [state how labels were produced].
- **External validity.** The pilot uses three nodes in one area during one season, so results may not generalise to other soils, varieties, water-management regimes or seasons.
- **Construct validity.** Prediction accuracy against reference labels is a proxy for good irrigation; measured water use and crop condition are needed to confirm real benefit.
- **Conclusion validity.** A small number of plots limits statistical power, so comparisons should be interpreted cautiously.
- **Operational limits.** Seasonal data drift, sensor failure, connectivity loss and low farmer adoption remain risks (Appendix A).

7.6 Implications

For farmers, the component offers a simple recommendation rather than raw sensor values, which can reduce water wastage and the effort of deciding when to irrigate. For agricultural officers and RRDI stakeholders, the irrigation history and field data provide evidence for advisory work and research. For the platform, shared irrigation context improves the yield, disease and pest components, illustrating the value of interoperable services over standalone tools.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The target market is smallholder paddy farmers and farmer clusters that need a low-cost irrigation advisory tool rather than an expensive full-automation package. The component is positioned as a practical decision-support service that can be deployed first as a pilot bundle and later as a subscription-supported service. Its main value lies in localised stage-aware scheduling, bilingual farmer communication and integration with yield, disease and pest analytics.

8.2 Market and competitive position

Unlike generic irrigation timers, the component is designed around Sri Lankan paddy seasons and common smallholder constraints. Compared with flood irrigation and canal rotations it adds field-level precision; compared with threshold systems it adds stage-aware prediction; and compared with generic ML schedulers it is tailored to local conditions and delivered in Sinhala and English (Table 2.2). Hardware costs are kept low by using off-the-shelf ESP32 and DFRobot components.

8.3 Business or sustainability model

A feasible rollout path is to pilot the component with a small number of fields, demonstrate measurable savings, and then partner with farmer organisations, agrarian service centres or agritech integrators for cluster-based adoption. A monthly subscription of around LKR 500 can support notifications, cloud synchronisation and model maintenance after the initial pilot kit is deployed (Table 8.1). At platform level, access is tiered: a free tier with basic image-based disease detection and weather alerts; a subscription tier with predictive risk forecasts, stage-aware irrigation schedules and eco-friendly pest mitigation plans; and an institutional tier with district-level dashboards for cooperatives, extension services and RRDI researchers.

Table 8.1. Indicative pricing

Item	Value	Purpose

Item	Value	Purpose
Pilot sensor starter bundle	LKR 75,000	Three field-ready node kits for basic spatial coverage
Cloud sync and alert service	LKR 5,000	Firebase usage, testing and notification support during pilot
Indicative farmer subscription after pilot	LKR 500 / month	Supports app access, synchronisation and model updates

The pilot budget is intentionally compact because the objective is to validate the component rather than deploy a large commercial network (Table 8.2).

Table 8.2. Pilot budget

Item	Qty	Unit cost (LKR)	Total (LKR)	Justification
Sensor node bundle (probes, controller, enclosure)	3	25,000	75,000	Three units allow comparison across field points
Cloud / Firebase / notifications	-	5,000	5,000	Real-time sync and alert delivery during the pilot
Total			**80,000**	Compact validation budget for the component pilot

8.4 Scalability intellectual property and compliance

The cloud tier is designed to scale from a pilot to thousands of smallholder farms through containerised microservices and a time-series data store, while hardware can be funded in stages or bought through cluster purchasing by farmer cooperatives. Intellectual property in the models, pipeline and application belongs within the project and institutional framework of SLIIT [confirm the IP position with the supervisor]. Compliance requirements include informed consent, anonymised field identifiers, authenticated and role-based access, and data handling consistent with Sri Lanka's Personal Data Protection Act, No. 9 of 2022.

8.5 Contribution to the Sustainable Development Goals

The component contributes to several Sustainable Development Goals (SDGs). It supports SDG 2 (Zero Hunger) by protecting smallholder yield from water stress; SDG 6 (Clean Water and Sanitation) through more efficient water use; SDG 12 (Responsible Consumption and Production) by reducing resource waste and unnecessary chemical use through better field conditions; and SDG 13 (Climate Action) by helping farmers adapt to erratic monsoon patterns.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis addressed the lack of an affordable, stage-aware irrigation scheduler for Sri Lankan smallholder paddy. It designed and built the Paddy Irrigation Optimization component, which combines low-cost soil and weather sensing, growth-stage features and a scikit-learn scheduler, and delivers recommendations through cloud APIs to a bilingual mobile application with manual override and safe fallback alerts. The component is integrated with the yield, disease and pest components of the IoT and AI-based smart agriculture platform. [Add a concluding statement on the measured results once Chapter 6 is complete.]

9.2 Achievement of objectives

Table 9.1. Achievement of the specific objectives

ID	Objective	Evidence	Status
SO1	Deploy and calibrate field sensors	Stable pilot data stream (Section 5.5)	[insert]
SO2	Develop a stage-aware ML model	Validation results (Table 6.1)	[insert]
SO3	Integrate with cloud APIs and the mobile app	Integration and testing (Sections 5.3, 5.4, 6.4)	[insert]
SO4	Validate in field conditions	Water savings and user feedback (Sections 6.1, 6.3)	[insert]

9.3 Individual research and product contributions

The individual research contribution is a stage-aware, locally validated approach to irrigation scheduling for Sri Lankan paddy and an evaluation design that compares it with reactive threshold logic and current farmer practice. The product contribution is the sensing-to-recommendation pipeline, the bilingual application with manual override, and the interfaces that connect irrigation to the other platform components.

9.4 Future directions

- Extend the pilot to more fields, soil types, rice varieties and both seasons to test generalisation and retrain the model on seasonal data.
- Add evapotranspiration-based and satellite or forecast-derived features to complement point sensors.
- Move from advisory recommendations to supervised valve actuation once reliability is demonstrated.
- Use farmer feedback and logged outcomes to refine the model continuously, and explore adaptive learning methods.
- Deepen the integration with disease and pest components, for example by coordinating irrigation with spray windows.
- Adapt the pipeline to other crops and to cooperative-level dashboards for extension services.

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GLOSSARY

Alternate wetting and drying - A water-saving practice in which the paddy field is allowed to dry to a set level between irrigations instead of staying continuously flooded.

Edge node - A low-cost field device (microcontroller with sensors) that captures and pre-validates readings before sending them to the cloud.

Electrical conductivity (EC) - A measure of how well soil water conducts electricity, used as an indicator of dissolved salts and nutrients.

Flood irrigation - Irrigation by keeping or repeatedly flooding the field with water, common in paddy cultivation.

Grain filling - The late growth stage in which grains accumulate starch and reach maturity.

Maha season - The main cultivation season in Sri Lanka, associated with the north-east monsoon.

Manual override - A control that lets the farmer accept, delay or reject a system recommendation.

Panicle initiation - The growth stage in which the rice panicle begins to form inside the plant.

Stage-aware scheduling - Irrigation scheduling that takes the current crop growth stage into account when judging water demand.

Tillering - The early vegetative stage in which the rice plant produces additional shoots (tillers).

Yala season - The minor cultivation season in Sri Lanka, from roughly April to September.

APPENDICES

Appendix A Data governance and risk register

This appendix gives the data-governance notes and the risk register supporting Sections 3.6 and 7.5.

Data governance notes. Farmer participation is voluntary and based on informed consent. Field identifiers in the dataset are coded to avoid exposing personally identifiable information. Sensor data, weather feeds and mobile interactions are used only for research and pilot evaluation within the project scope, and administrative access to the cloud database is restricted to authorised team members.

Table A.1. Risk register

Risk	Level	Impact	Mitigation
Sensor failure	Medium	Missing or incorrect recommendations	Keep backup probes and trigger manual-inspection alerts
Connectivity loss	Medium	Delayed synchronisation to cloud	Buffer readings locally and retry sync when the network returns
Seasonal data drift	High	Reduced model accuracy across Maha and Yala	Retrain with seasonal data and monitor validation metrics
Low farmer adoption	Medium	Limited practical impact of the pilot	Use bilingual alerts, simple guidance and field demonstrations

Appendix B Academic integrity and AI use disclosure

The author's use of generative AI and other automated tools is disclosed below in accordance with the module policy. The author remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions in this thesis.

Table B.1. Use of generative AI and automated tools

Tool	Purpose	Verification performed	Sections affected

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic)	Restructuring the author's proposal and final-presentation content into this thesis template; drafting prose, tables and figure scripts	[Author to confirm: facts checked against own project records; references checked; all measured results entered by the author]	Chapters 1-9, figures, appendices
scikit-learn, Python libraries	Model training and evaluation (software tools, not generative AI)	Results reproduced from versioned code and configuration	Chapters 3, 5 and 6

Appendix C Project schedule

The project plan covers requirements and design in the first quarter, deployment and observation of the pilot during the Yala season, and the first validation cycle by July 2026. Milestones are the prototype in April, completion of field validation in June and final evaluation in July.

Table C.1. Work breakdown structure

ID	Task	Deliverable	Duration
1.0	Requirements and system design	Reviewed requirements and architecture draft	4 weeks
2.0	Sensor setup and field deployment	Calibrated nodes and field data stream	4 weeks
3.0	Data preprocessing and feature engineering	Cleaned dataset and modelling pipeline	3 weeks
4.0	ML model development	Validated scheduler model and comparison report	6 weeks
5.0	App and API integration	Operational alerts and data synchronisation	4 weeks
6.0	Field validation	Pilot results with measured savings and feedback	4 weeks
7.0	Documentation and revisions	Updated report and evaluation summary	3 weeks

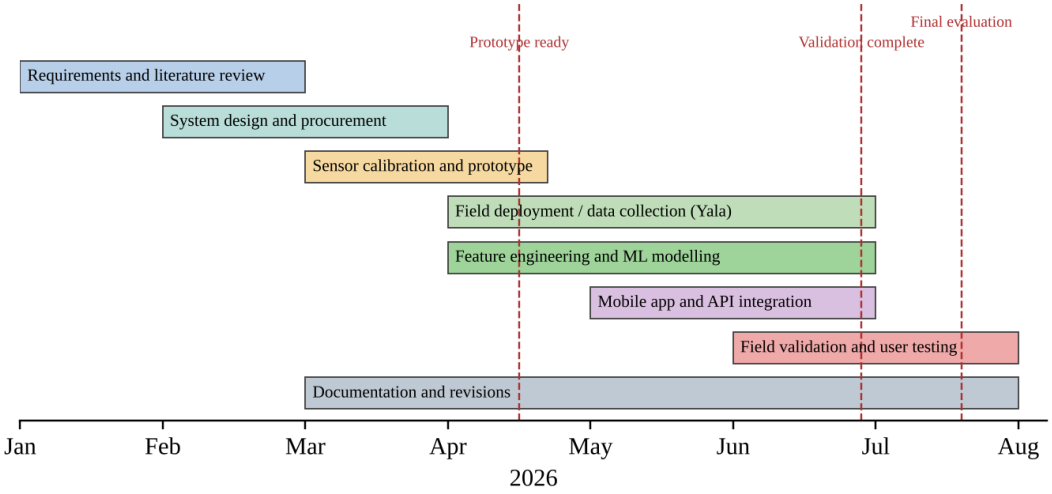


Figure C.1. Project Gantt chart for the paddy irrigation optimization component